# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME: INTERNET PROGRAMMING**
**COURSE CODE: CSA4303**
**STUDENT NAME: M.Yugandhar Reddy**
**REG. NO.: 192372016**
**MARKS: 76**

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**COURSE OUTCOME COVERED**

**CO-1:** Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

**Weightage: 10%**

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**QUESTIONS:** **Total Marks: 10**

## Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols. The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

## Data Snippets Provided

### Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

### Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

### Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

### Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

### Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

## Questions and Answers:

**1. Select the correct HTML tag for video embedding and justify your answer.**

**Answer:** Option B: `<video src="video.mp4" controls></video>`

**Justification:**

- **HTML5 native element:** `<video>` is the only standard element for embedding video; the browser plays it without plug-ins such as Flash.
- **controls attribute:** shows the built-in play, pause, volume, seek and fullscreen buttons, so visitors can control the promotional video.
- **Why the others are wrong:** `<media>`, `<movie>` and `<vid>` do not exist in the HTML specification, so browsers ignore them and nothing plays.
- **Good practice:** add width/height, a poster image and fallback text between the tags, or use `<source>` for several formats (mp4, webm).

**2. Identify the appropriate HTTP method for form submission and explain why.**

**Answer:** Option B: POST

**Justification:**

- **Data in the request body:** POST sends registration data (name, email, password) in the body, so it does not appear in the URL, browser history, bookmarks or server logs.
- **GET** appends data to the URL (visible and limited in length), so it is unsuitable for sensitive data; it is meant for retrieving data.
- **POST** is meant for creating or submitting data to the server, which is what registration does, and it has no practical size limit.
- **PUT** replaces an existing resource and **DELETE** removes one; they are used in REST APIs, and plain HTML forms support only GET and POST.
- **Security note:** POST alone does not encrypt data; it must be used together with HTTPS.

**3. Choose the proper semantic element for navigation and justify its usage.**

**Answer:** Option B: `<nav>`

**Justification:**

- **Meaning:** `<nav>` marks a block of major navigation links, such as the main menu.
- **Accessibility:** screen readers announce it as a navigation landmark so users can jump straight to it.
- **SEO and maintainability:** search engines and developers understand the page structure more easily.
- **Others:** `<div>` has no meaning, `<section>` groups thematic content, and `<header>` holds introductory content (it may contain a `<nav>`).

**4. Identify the correct XML syntax and explain the error in other options.**

**Answer:** Option B: `<note><to>User</to><from>Admin</from></note>`

**Justification:**

- **Why B is correct:** it is well-formed: a single root element, every start tag has a matching end tag, and elements are properly nested.
- **A is wrong:** the root element `<note>` is never closed (missing `</note>`).
- **C is wrong:** `<to>` is closed with `</from>`, a mismatched tag, which breaks nesting.
- **D is wrong:** the closing tag `</note` is missing its `>`, so the element is malformed. The attribute form `to="User"` is fine otherwise.

**5. Choose the correct protocol for web communication and justify your choice.**

**Answer:** Option B: HTTP (HTTPS for secure communication)

**Justification:**

- **Purpose:** HTTP (HyperText Transfer Protocol) is the application-layer protocol browsers and servers use to request and deliver web pages and resources, over TCP port 80.
- **Request-response model:** the client sends a request (GET, POST...) and the server returns a response with a status code and content; it is stateless.
- **HTTPS:** HTTP over TLS (port 443) encrypts traffic, which the registration form requires.
- **Others:** FTP transfers files, SMTP sends email, and SNMP manages network devices; none is used for delivering web pages.

## Code Along: My First HTML Page with CSS

A small page that applies all the correct answers above (video, POST form over HTTPS, semantic nav) and styles it with CSS.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>My First HTML Page</title>
  <style>
    body   { font-family: Arial, sans-serif; margin: 0; }
    header { background: #1F3864; color: #fff; padding: 16px; }
    nav a  { color: #fff; margin-right: 14px; text-decoration: none; }
    main   { padding: 16px; }
    video  { width: 100%; max-width: 480px; }
    form   { display: grid; gap: 8px; max-width: 320px; }
  </style>
</head>
<body>
  <header>
    <h1>StartUp App</h1>
    <nav>
      <a href="#home">Home</a>
      <a href="#video">Video</a>
      <a href="#register">Register</a>
    </nav>
  </header>
  <main>
    <section id="video">
      <h2>Promo Video</h2>
      <video src="video.mp4" controls>Your browser does not support video.</video>
    </section>
    <section id="register">
      <h2>Register</h2>
      <form action="https://example.com/register" method="post">
        <input type="text" name="name" placeholder="Name" required>
        <input type="email" name="email" placeholder="Email" required>
        <input type="password" name="password" placeholder="Password" required>
        <button type="submit">Sign up</button>
      </form>
    </section>
  </main>
</body>
</html>
```

The XML configuration file, corrected:

```
<?xml version="1.0" encoding="UTF-8"?>
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

### Code of Conduct:

I **M.Yugandhar Reddy / 192372016** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ***M.Yugandhar Reddy***

## RUBRICS FOR CALCULATION:

| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning | Gained Marks |
|---|---|---|---|---|---|---|
| **Understanding of Case** | 25 | Thorough understanding; clearly identifies all key issues | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly | **19** |
| **Application of Concepts** | 25 | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps | Limited application of concepts | Fails to apply relevant concepts | **19** |
| **Analysis** | 20 | In-depth analysis with strong reasoning and insights | Good analysis with logical reasoning | Basic analysis with limited depth | Weak or no analysis; lacks reasoning | **15** |
| **Solution Design** | 20 | Innovative, feasible solutions with strong justification | Logical solutions with adequate justification | Basic solutions with weak justification | Incorrect or no solution provided | **15** |
| **Clarity** | 10 | Clear, well-structured, and easy to understand | Organized and understandable presentation | Limited clarity and structure | Poor clarity; difficult to understand | **8** |
| **Total** | **100** | | | | | **76** |